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create table clean_transactions
(transaction_id varchar,transaction_date varchar, customer_id varchar,
product_name varchar,quantity varchar,price varchar
,payment_method varchar , transaction_status varchar)

select * from clean_transactions

DROP TABLE IF EXISTS clean_transactions;

CREATE TABLE clean_transactions (
    transaction_id      SERIAL PRIMARY KEY,
    transaction_date    DATE,
    customer_id         VARCHAR,
    product_name        VARCHAR,
    quantity            INT,
    price               NUMERIC,
    payment_method      VARCHAR,
    transaction_status  VARCHAR
);

CREATE TABLE staging_transactions (
    transaction_id      VARCHAR(255),
    transaction_date    VARCHAR(255),
    customer_id         VARCHAR(255),
    product_name        VARCHAR(255),
    quantity            VARCHAR(255),
    price               VARCHAR(255),
    payment_method      VARCHAR(255),
    transaction_status  VARCHAR(255)
);

select * from staging_transactions

CREATE TEMP TABLE temp_formatted AS
SELECT
    NULLIF(TRIM(transaction_date), '') AS clean_date,
    TRIM(customer_id) AS customer_id,
    TRIM(product_name) AS product_name,
    NULLIF(TRIM(quantity), '') AS clean_qty,
    NULLIF(TRIM(REPLACE(REPLACE(REPLACE(price, '$', ''), '(', '-'), ')', '')), '') AS clean_price,
    LOWER(TRIM(payment_method)) AS payment_method,
    INITCAP(TRIM(transaction_status)) AS transaction_status
FROM staging_transactions;

INSERT INTO clean_transactions (
    transaction_date,
    customer_id,
    product_name,
    quantity,
    price,
    payment_method,
    transaction_status
)
SELECT
    clean_date::DATE,
    customer_id,
    product_name,
    clean_qty::INT,
    clean_price::NUMERIC(10,2),
    payment_method,
    transaction_status

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DROP TABLE IF EXISTS temp_formatted;

CREATE TEMP TABLE temp_formatted AS
SELECT
    CASE
        -- Check if string matches YYYY-MM-DD pattern and isn't February 30/31 or invalid day
        WHEN TRIM(transaction_date) ~ '^d{4}-d{2}-d{2}$'
            AND TRIM(transaction_date) NOT LIKE '%-02-29'
            AND TRIM(transaction_date) NOT LIKE '%-02-30'
            AND TRIM(transaction_date) NOT LIKE '%-02-31'
            AND TRIM(transaction_date) NOT LIKE '%-04-31'
            AND TRIM(transaction_date) NOT LIKE '%-06-31'
            AND TRIM(transaction_date) NOT LIKE '%-09-31'
            AND TRIM(transaction_date) NOT LIKE '%-11-31'
        THEN TRIM(transaction_date)::DATE
        ELSE NULL
    END AS clean_date,

    TRIM(customer_id) AS customer_id,
    TRIM(product_name) AS product_name,

    -- Keep valid integer strings only
    CASE
        WHEN TRIM(quantity) ~ '^-\d+$' THEN TRIM(quantity)::INT
        ELSE NULL
    END AS clean_qty,

    -- Strip $ and formatting, convert to numeric safely
    CASE
        WHEN NULLIF(TRIM(REPLACE(REPLACE(REPLACE(price, '$', ''), '(', '-'), ')', '')), '') ~ '^-\d+(\.\d+)?$'
        THEN TRIM(REPLACE(REPLACE(REPLACE(price, '$', ''), '(', '-'), ')', ''))::NUMERIC(10,2)
        ELSE NULL
    END AS clean_price,

    LOWER(TRIM(payment_method)) AS payment_method,
    INITCAP(TRIM(transaction_status)) AS transaction_status
FROM staging_transactions;

CREATE OR REPLACE FUNCTION safe_cast_date(val TEXT)
RETURNS DATE AS $$
BEGIN
    RETURN val::DATE;
EXCEPTION WHEN OTHERS THEN
    RETURN NULL;
END;
$$ LANGUAGE plpgsql;

DROP TABLE IF EXISTS temp_formatted;

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CREATE TEMP TABLE temp_formatted AS
SELECT
    safe_cast_date(TRIM(transaction_date)) AS clean_date,
    TRIM(customer_id) AS customer_id,
    TRIM(product_name) AS product_name,

    CASE
        WHEN TRIM(quantity) ~ '^-\d+$' THEN TRIM(quantity)::INT
        ELSE NULL
    END AS clean_qty,

    CASE
        WHEN NULLIF(TRIM(REPLACE(REPLACE(REPLACE(price, '$', ''), '(', '-'), ')', '')), '') ~ '^-\d+(\.\d+)?$'
        THEN TRIM(REPLACE(REPLACE(REPLACE(price, '$', ''), '(', '-'), ')', ''))::NUMERIC(10,2)
        ELSE NULL
    END AS clean_price,

    LOWER(TRIM(payment_method)) AS payment_method,
    INITCAP(TRIM(transaction_status)) AS transaction_status
FROM staging_transactions;

CREATE OR REPLACE FUNCTION safe_cast_date(val TEXT)
RETURNS DATE AS $$
BEGIN
    RETURN val::DATE;
EXCEPTION WHEN OTHERS THEN
    RETURN NULL;
END;
$$ LANGUAGE plpgsql;

DROP TABLE IF EXISTS temp_formatted;

CREATE TEMP TABLE temp_formatted AS
SELECT
    safe_cast_date(TRIM(transaction_date)) AS clean_date,
    TRIM(customer_id) AS customer_id,
    TRIM(product_name) AS product_name,

    CASE
        WHEN TRIM(quantity) ~ '^-\d+$' THEN TRIM(quantity)::INT
        ELSE NULL
    END AS clean_qty,

    CASE
        WHEN NULLIF(TRIM(REPLACE(REPLACE(REPLACE(price, '$', ''), '(', '-'), ')', '')), '') ~ '^-\d+(\.\d+)?$'
        THEN TRIM(REPLACE(REPLACE(REPLACE(price, '$', ''), '(', '-'), ')', ''))::NUMERIC(10,2)
        ELSE NULL
    END AS clean_price,

    LOWER(TRIM(payment_method)) AS payment_method,
    INITCAP(TRIM(transaction_status)) AS transaction_status
FROM staging_transactions;

INSERT INTO clean_transactions (
    transaction_date,
    customer_id,
    product_name,
    quantity,
    price,
    payment_method,
    transaction_status
)
SELECT
    clean_date,
    customer_id,
    product_name,

```